

The output, employment, and interest rate effects of government consumption*

S. Rao Aiyagari and Lawrence J. Christiano

Federal Reserve Bank of Minneapolis, Minneapolis, MN 55480, USA

Martin Eichenbaum

Northwestern University, Evanston, IL 60201, USA

Federal Reserve Bank of Chicago, Chicago, IL 60604, USA

National Bureau of Economic Research, Cambridge, MA 02138, USA

Received January 1991, final version received July 1992

We study the impact on aggregate variables of changes in government consumption using the neoclassical stochastic growth model. We show, theoretically, that the impact on output and employment of a persistent change in government consumption exceeds that of a temporary change. We also show that, in principle, there can be an analog to the Keynesian multiplier in the neoclassical growth model. Finally, in an empirically plausible version of the model, we show that the interest rate impact of a persistent government consumption shock exceeds that of a temporary one. Our results provide counterexamples to existing claims in the literature.

1. Introduction

In this paper, we investigate the impact of changes in government consumption on aggregate employment, output, and interest rates. We proceed at both a theoretical and a quantitative level. These issues have been previously analyzed by Hall (1980) and Barro (1981, 1987), ostensibly using the standard neoclassical growth model. However, our analysis indicates that the neoclassical growth model makes predictions about the impact of government consumption

Correspondence to: S. Rao Aiyagari, Research Department, Federal Reserve Bank of Minneapolis, P.O. Box 291, Minneapolis, MN 55480-0291, USA.

*We are grateful to Lars Hansen for helpful comments. Christiano and Eichenbaum acknowledge research support from the National Science Foundation. The views expressed herein are those of the authors and not necessarily those of the Federal Reserve Bank of Minneapolis, the Federal Reserve Bank of Chicago, or the Federal Reserve System.

which are, in many ways, fundamentally different from those claimed by Hall and Barro.

It is convenient to begin by summarizing those claims. First, according to Hall (1980) and Barro (1981, 1987), both transient and persistent increases in government consumption ought to increase aggregate output and hours worked. Second, Hall (1980) argues that the employment and output effects of persistent increases in government consumption are smaller than those of transient increases.¹ Third, according to Barro (1987), the ratio of the change in contemporaneous and steady-state output to changes in government consumption must be less than one; i.e., there can be no analog to the Keynesian multiplier in the neoclassical growth model. Fourth, Barro (1981, 1987) argues that a transient increase in government consumption ought to impact positively on the interest rate, while a permanent increase ought to leave the interest rate unchanged.

While our theoretical analysis leads us to conclude, as do Hall (1980) and Barro (1981, 1987), that transient as well as persistent increases in government consumption do generate increases in output and employment, we differ along other important dimensions. First, we demonstrate analytically that under standard assumptions the employment and output effects of persistent increases in government consumption always *exceed* those of temporary increases.² From this perspective, the more persistent is government consumption, the more likely it is to be an important impulse to business cycle fluctuations. Second, we use a parametric version of our model to show that the steady-state response of output to a permanent one-unit change in government consumption can exceed unity. Further, the contemporaneous response of output to a one-unit change in government consumption can exceed unity if that change is sufficiently persistent. Thus, in principle, there *can* be an analog to the Keynesian multiplier in conventional neoclassical models. Third, we show that, in our parametric model, an increase in government consumption causes the real interest rate to rise, regardless of whether the change in government consumption is temporary or persistent. Indeed, for that model, we find that a persistent

¹Barro (1981) seems to suggest this as well. In particular, he writes that 'under plausible conditions, the temporary case involves an output response that is positive, less than one-to-one with the change in government purchases, and larger than that generated by an equal-sized, but permanent, shift in purchases' [Barro (1981, p. 1086)]. However, he comments on the relative output effects of temporary and permanent changes in government purchases only in the abstract of the paper and in the section on optimal, non-lump-sum taxes – a case we do not consider. Therefore, it is not clear whether the previous statement is meant to apply both to lump-sum and optimal non-lump-sum taxation situations or only to the latter.

²This verifies conjectures in Baxter and King (1988) and Christiano and Eichenbaum (1992) that such a result might hold for a reasonably general class of models. See also Barro and King (1984, p. 831, n. 13), who suggest that under lump-sum taxes a permanent change in government purchases would have a stronger effect than a temporary change in government purchases because 'investment tends to decline more when the change in purchases is temporary'.

increase in government consumption has a *larger* impact on the interest rate than does a transient increase.

In the remainder of this section we provide some intuition for the differences between our results and those in the existing literature.

1.1. The relative impact of temporary versus persistent government consumption shocks

The basic argument underlying Hall's (1980) conclusion about the relative impact of persistent and temporary changes in government consumption can be summarized as follows. A temporary increase of one unit in government consumption reduces consumers' permanent incomes by much less than one unit (approximately zero). Consequently, the demand for private consumption remains (roughly) unchanged, so that, for a given interest rate, net aggregate demand increases by one unit. Since there has been no change in labor suppliers' permanent incomes, other things equal, when viewed as a function of the interest rate, the aggregate supply curve of output does not shift. Consequently, real interest rates must rise to clear the commodity market. This transient increase in the interest rate induces a positive labor supply response for intertemporal substitution reasons.

In contrast, a permanent increase in government consumption by one unit reduces consumers' permanent incomes by one unit. Other things equal, this induces a fall in the demand for private consumption by one unit. With no net change in aggregate demand, there is no need for any offsetting effects on interest rates. Consequently, equilibrium output and hours worked do not change.

The basic problem with Hall's (1980) argument is that it makes inconsistent assumptions regarding the income effect on leisure. An implicit assumption of his analysis of a permanent change in government consumption is that there is no income effect on leisure. But this cannot also be a maintained assumption of his analysis of a temporary change in government consumption. This is because, under that assumption, a temporary change in government consumption *also* has no impact on equilibrium output and employment.³

In fact, with a positive income effect on leisure, permanent changes in government consumption always have effects on employment and output that are larger than the effects of transitory changes. We show this by decomposing

³This follows from two observations. First, if the income effect on leisure is zero, then the marginal rate of substitution between labor and consumption depends only on leisure. Therefore, households' labor supply depends only on the current real wage rate – interest rate considerations play no role in determining labor supply. Second, the marginal productivity schedule of labor, which determines the demand for labor, is fixed by the current stock of capital and the level of technology. Taken together, these two observations imply that equilibrium output and employment do not depend on the contemporaneous level of government consumption.

the effect on hours worked of a change in government consumption as follows: (a) the direct effect of the change in government consumption on hours worked, with private investment held constant, (b) the effect on investment of the change in government consumption, and (c) the indirect effect of government consumption on hours worked that occurs via the change in investment.

As it turns out, in (a) and (c), increases in government consumption and investment can be viewed as exogenous reductions in income which raise hours worked. However, with respect to (b), a transient increase in government consumption *reduces* investment, reflecting agents' desire to smooth consumption over time. But a *persistent* increase in government consumption either *increases* investment or does not reduce it by as much as in the transient case. Basically, this is because a permanent increase in government consumption raises the steady-state level of capital. It is the indirect effect of government that accounts for the fact that persistent changes in government consumption generate *larger* contemporaneous effects on output and employment than transient changes.

1.2. *The multiplier*

The basic argument underlying Barro's (1981, 1987) conclusion that contemporaneous and steady-state output increase by less than a given permanent increase in government consumption can be summarized as follows. The negative wealth effect associated with a permanent increase in government consumption leads to a reduction in private consumption. At the same time, investment does not change, because according to Barro's analysis, interest rates do not change. Since aggregate output equals the sum of private consumption, investment, and government consumption, the increase in output must be less than the increase in government consumption.

The key problem with this argument lies in the assertion that investment does not change in response to a permanent increase in government consumption. As long as leisure is a superior good and the marginal product of capital depends on hours worked, investment must change. To see why, it is convenient to consider the nonstochastic case. In standard versions of the neoclassical growth model, the steady-state capital/hours worked ratio is determined by agents' subjective discount rate and is not affected by government consumption. But given a positive income effect on leisure, steady-state hours worked increase in response to a permanent increase in government consumption. It follows that the steady-state stock of capital and investment must also increase. Given the increase in steady-state investment, there is no longer any reason for the steady-state multiplier to be less than one. Similar considerations imply that the contemporaneous output effect of a permanent change in government consumption can also exceed one.

1.3. *The interest rate and government consumption*

According to Barro (1981, 1987), permanent increases in government consumption ought to have no impact on the interest rate. To see why this result does not hold in general, it is again useful to consider the nonstochastic case. With a positive income effect on leisure, permanent increases in government consumption drive up the steady-state stock of capital and hours worked. Under standard assumptions, the interest rate is above its steady-state value whenever the capital stock is below its steady-state value. Moreover, convergence is monotonic, with the interest rate eventually returning to its unchanged steady-state value [Aiyagari (1988, prop. 3)]. Consequently, a permanent increase in government consumption induces a temporary increase in the interest rate. Indeed, in our parametric model, we find that this increase is actually larger than that which would result from a temporary change in government consumption.⁴

The remainder of this paper is organized as follows. In section 2 we present our basic model and analytical results. We refer the reader to Aiyagari et al. (1990) for proofs. Section 3 contains our numerical results. Finally, section 4 contains some concluding remarks.

2. *Dynamic effects of government consumption: Theoretical results*

In this section we analyze a variant of the neoclassical growth model modified to include elastic labor supply and government consumption. Our main result is that the contemporaneous effect on employment and output of a persistent increase in government consumption always exceeds the corresponding effect of a temporary increase in government consumption. The purpose of this section is to provide an intuitive description of the argument.

2.1. *The model*

We assume that the competitive equilibrium allocation can be represented as the solution to the following planning problem. Maximize

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t, n_t), \quad 0 < \beta < 1, \quad (1)$$

⁴We wish to emphasize that this result is by no means a general one. There exist alternative preference and technology assumptions for the one-sector growth model in which temporary increases in government consumption have bigger impacts on the interest rate than permanent changes. To see this, suppose the economy is in nonstochastic steady state and there is no income effect on leisure. Then a permanent increase in government consumption that is met by an equal

subject to

$$c_t + g_t + k_{t+1} \leq f(k_t, n_t), \quad (2)$$

and $0 \leq n_t \leq N$, $c_t \geq 0$, $k_{t+1} \geq 0$, $k_0 > 0$ given by choice of contingency plans for n_t , c_t , and k_{t+1} . Here, c_t , g_t , and k_t denote private consumption, public consumption, and the beginning-of-period public plus private stock of capital, respectively. Also, n_t denotes hours worked and N is the total time endowment. The function $f(k_t, n_t)$ relates date t gross output and the undepreciated part of capital to the factors of production. Our assumptions on preferences and technology [Aiyagari et al. (1990)] are entirely standard. The key assumption on preferences is that leisure is a superior good.

We assume that government consumption, g_t , evolves according to

$$\begin{aligned} g_t &= G(g_t^T + g_t^P), \\ -\infty &< g_{\min}^T \leq g_t^T \leq g_{\max}^T < \infty, \\ -\infty &< g_{\min}^P \leq g_t^P \leq g_{\max}^P < \infty. \end{aligned} \quad (3)$$

Here, $G(\cdot) > 0$ is a strictly increasing function while g_t^T and g_t^P are the zero mean transitory and persistent components of government consumption, respectively. Throughout, we assume that g_t^T is iid over time and that g_t^T and g_t^P are mutually independent. Further, g_t^P is a Markov process with conditional cumulative distribution function $\Psi(g^{P'} | g^P) = \text{Prob}[g_{t+1}^P \leq g^{P'} | g_t^P = g^P]$. We make the following assumption for Ψ :

Assumption E1. Ψ is decreasing in g^P .⁵

Assumption E1 formalizes our notion of persistence. Intuitively, g_t^P is persistent if higher values of g_t^P make higher values of g_{t+1}^P more likely.

2.2. Government consumption and equilibrium employment and output

We discuss the qualitative features of the solution to our planning problem using standard dynamic programming methods. Bellman's equation for the

decrease in private consumption and no change in hours worked is both feasible and optimal. But this means that the interest rate is left unchanged by the permanent increase in government consumption. If the increase in government consumption is temporary, the proposed consumption and hours worked response would be feasible, but not optimal. In fact, investment would fall and the interest rate would rise.

⁵Formally, that Ψ is decreasing in g^P means that if $g_2^P > g_1^P$, then $\Psi(g^{P'} | g_1^P) \geq \Psi(g^{P'} | g_2^P)$ for all $g^{P'}$, with the inequality being strict for some $g^{P'}$.

planning problem is

$$v(k, g^P, g^T) = \max_{c, k', n \in A(k, g)} \{u(c, n) + E[\beta v(k', g^{P'}, g^{T'}) | g^P]\}, \quad (4)$$

where $k \in [\underline{k}, \bar{k}]$ and

$$A(k, g) = \{c, k', n : c \geq 0, 0 \leq n \leq N, k' \geq \underline{k}, c + k' + g \leq f(k, n)\}. \quad (5)$$

In (4)–(5), time subscripts have been deleted and the primes (') denote next period's value. The $\underline{k}$ and $\bar{k}$ are the minimum and maximum sustainable capital stocks, respectively. The v is the optimal value function. The contingency plans for c , n , and k' that we seek are the unique plans which attain the maximum in (4).

For our purposes, it is convenient to rewrite (4)–(5) as

$$v(k, g^P, g^T) = \max_{\underline{k} \leq k' \leq f(k, N) - g} \left\{ \left[\max_{n, c \in B(k, k' + g)} u(c, n) \right] + E[\beta v(k', g^{P'}, g^{T'}) | g^P] \right\}, \quad (6)$$

where

$$B(k, k' + g) = \{c, n : 0 \leq n \leq N, 0 \leq c \leq f(k, n) - (g + k')\}, \quad (7)$$

for $k \in [\underline{k}, \bar{k}]$, $k' \in [\underline{k}, f(k, N) - g]$. Let the function defined in the first brackets of (6) be denoted

$$W(k, k' + g) = \max_{n, c \in B(k, k' + g)} u(c, n). \quad (8)$$

There exist unique solutions,

$$n = h(k, k' + g), \quad c = q(k, k' + g), \quad (9)$$

to the maximization problem in (8). With this notation, (4) can be written as

$$v(k, g^P, g^T) = \max_{\underline{k} \leq k' \leq f(k, N) - g} \{W(k, k' + g) + E[\beta v(k', g^{P'}, g^{T'}) | g^P]\}. \quad (10)$$

Thus, problem (4) is broken into two parts, a purely static part, (8), and a dynamic part, (10). This way of splitting the problem corresponds to our description in the introduction of the direct and indirect effects of movements in

g^T . Problem (8) captures the direct effects of g and k' on (c, n) , whereas problem (10) captures the indirect effects of g on (c, n) via k' .

Notice that the distinction between transient and persistent changes in g enters *solely* via the indirect effect in problem (10). One way to see this is to decompose the response of n to a change in g as

$$\frac{dn}{dg} = \frac{\partial h}{\partial g} + \frac{\partial h}{\partial k'} \frac{dk'}{dg} = \frac{\partial h}{\partial g} \left[1 + \frac{dk'}{dg} \right], \quad (11)$$

where h is as defined in (9).⁶ The expression after the second equality in (11) reflects the symmetry with which g and k' enter the h function. Because g^P and g^T enter the static problem's constraint set, B , in a symmetric way, it follows that $\partial h / \partial g$ does not depend on whether the change in g reflects a change in g^P or g^T . However, because a change in g^P influences the distribution of g' while a change in g^T does not, no such symmetry exists for dk' / dg . Hence, our problem reduces to understanding the relative magnitudes of $(dk' / dg)^P$ and $(dk' / dg)^T$.

2.3. The static problem

It is easy to see that h is strictly increasing in $k' + g$ by considering the planner's static consumption/leisure choice problem. Here, an increase in $k' + g$ corresponds to a parallel downward shift in the planner's resource constraint set. Since leisure is a superior good, the ensuing negative income effect generates an increase in hours worked. If the income effect on leisure is zero, then $\partial h / \partial g = 0$. This last result, in combination with (11), shows why the absence of an income effect on leisure cannot be used to rationalize standard arguments regarding the relative impacts of transient and persistent changes in government consumption on output and employment.

2.4. The dynamic problem

In this subsection we discuss the impact of transient and persistent changes in government consumption on investment. In particular, we analyze the relative magnitudes of $(dk' / dg)^T$ and $(dk' / dg)^P$. The optimal value of k' satisfies the following first-order condition implied by (10):

$$-W_k(k, k' + g) = E[\beta v_k(k', g^P, g^T) | g^P]. \quad (12)$$

⁶ Although our assumptions are sufficient to guarantee differentiability of h , we have not established differentiability of k' with respect to g , nor does our proof [in Aiyagari et al. (1990)] rely on differentiability of k' . Nevertheless, we find this notation useful for pedagogical purposes.

To understand how the optimal value of k' varies with changes in g^T and g^P , we plot $-W_{k'}(k, k' + g)$ and $E[\beta v_{k'}(k', g^P, g^T) | g^P]$ as functions of k' in fig. 1.

Consider first the graph of $E[\beta v_{k'}(k', g^P, g^T) | g^P]$, the (expected discounted) marginal benefit of an additional unit of k' (labelled B_1). The term $v_{k'}$, which equals $u_c(c', n')f_{k'}(k', n')$, is the value of an extra unit of capital next period. An increase in k' lowers the future marginal product of capital ($f_{k'}$). By raising c' , it also lowers the future marginal utility of consumption. Consequently, the marginal benefit curve is downward sloping. Fig. 1 also displays the graph of $-W_{k'}(k, k' + g)$, the marginal cost of next-period capital (labelled C_1). Since extra k' is obtained by forgoing current consumption, $-W_{k'}$ equals $u_c(c, n)$. From diminishing marginal utility of consumption, $-W_{k'}$ must be an increasing function of k' . The optimal value of k' is determined by the intersection of the marginal benefit and cost curves. In fig. 1, this value is denoted by k'_1 .

Now suppose there is a purely transitory increase in g ; i.e., $\Delta g^T > 0$. By construction, g^T is not an argument of $E[\beta v_{k'}(k', g^P, g^T) | g^P]$. Consequently, the marginal benefit curve does not shift. In contrast, g^T is an argument of $-W_{k'}(k, k' + g)$. Therefore, the marginal cost curve shifts to the left in response to an increase in g^T . In particular, the marginal cost curve shifts left by exactly the amount of the change in g since the marginal cost depends only on the sum $(k' + g)$. We label the new marginal cost curve C_2 and the new intersection point k'_2 . It is evident that $(dk'/dg)^T$ is negative and that $1 + (dk'/dg)^T$ is positive.

Now suppose that the increase in g is due to an increase in g^P . In contrast to the previous experiment, the marginal benefit curve is affected by this change

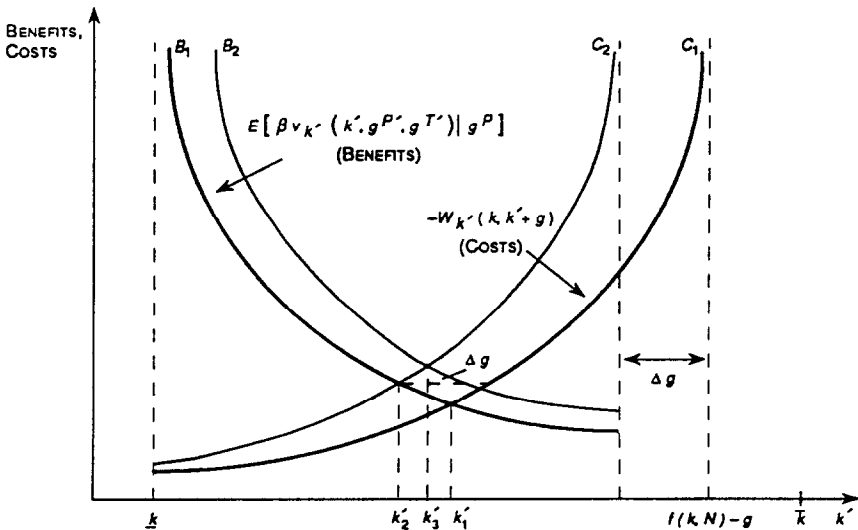


Fig. 1. Investment effects of transitory and persistent government consumption shocks.

because the conditional distribution of g^P depends on the value of g^P . To see the impact on $E\beta v_{k'}$ recall that $v_{k'}$ equals $u_c(c', n')f_{k'}(k', n')$. Other things equal, an increase in g^P lowers c' and, hence, raises u_c . It follows that $v_{k'}$ is an increasing function of g^P . By Assumption E1, higher values of g^P make higher values of g^P more likely. Consequently, $E[\beta v_{k'} | g^P]$ is an increasing function of g^P . [See Aiyagari et al. (1990) for a proof of this argument.]

In fig. 1, B_2 denotes the new marginal benefit curve corresponding to the higher value of g^P . The new marginal cost curve coincides with C_2 since g^P and g^T enter $W_{k'}$ in a symmetric fashion. The optimal value of k' under these circumstances is denoted by k'_3 . In general, one cannot determine the sign of $k'_3 - k'_1$. However, it *must* be the case that $k'_3 - k'_2$ is positive so that $(dk'/dg)^P > (dk'/dg)^T$. Therefore, $1 + (dk'/dg)^P > 1 + (dk'/dg)^T > 0$.

The previous result, in conjunction with (11), establishes the central result of our paper, namely, that *the contemporaneous impact on n (and, hence, on output) of a persistent increase in g exceeds the corresponding impact of a transient increase in g .*

3. Steady-state multipliers and the response of interest rates to government consumption

In this section we show that the nonstochastic steady-state impact of a persistent increase in government consumption can exceed unity. We also show that the impact on interest rates of a persistent change in government consumption can exceed that due to a temporary change in government consumption. We do these using a version of the parametric growth model considered by Christiano and Eichenbaum (1992). That model is a special case of the one analyzed in section 2, except that we allow for shocks to the aggregate production technology.⁷ Our numerical results were generated using values for the structural parameters estimated by Christiano and Eichenbaum (1992).

3.1. The parametric model

Our parametric model assumes that $u(c_t, n_t) = \ln(c_t) + \gamma \ln(N - n_t)$, where N denotes the time endowment, $f(\lambda_t, k_t, n_t) = \exp(-\theta \lambda_t) n_t^{(1-\theta)} k_t^\theta + (1-\delta) \exp(-\lambda_t) k_t$, and $G(g_t^P + g_t^T) = \bar{g} \exp(g_t^P + g_t^T)$. Our specification of f reflects the presence of a technology shock; the shock's time t growth rate we denote by λ_t .⁸ We assume that λ_t and g_t^T are mutually independent and

⁷It is straightforward to extend the results of section 2 to allow for this additional source of uncertainty.

⁸This specification actually corresponds to what Christiano and Eichenbaum (1992) refer to as the stationary version of the model and is obtained by transforming the nonstationary version of the model (which allows for growth) into a stationary dynamic programming planning problem.

independently distributed over time, with mean λ and 0, respectively. Regarding Ψ , we suppose that g_t^P has the AR(1) representation

$$g_t^P = \rho g_{t-1}^P + \mu_t, \quad |\rho| < 1, \quad (13)$$

where μ_t is fundamental for g_t^P and is distributed independently of λ_t .

3.2. Steady-state employment and output multipliers

Let $\eta_{n,g}$ and $\eta_{y,g}$ denote the elasticities of nonstochastic steady-state employment and output with respect to $\bar{g}$. The fact that the steady-state output/hours worked ratio is fixed by the discount rate implies that

$$\eta_{n,g} = \eta_{y,g} = \frac{dy_s}{d\bar{g}} \frac{\bar{g}}{y_s}. \quad (14)$$

It can be shown that

$$\frac{dy_s}{d\bar{g}} = \left\{ \frac{c_s + \bar{g}}{y_s} + \frac{c_s}{y_s} \frac{1}{[(N/n_s) - 1]} \right\}^{-1}. \quad (15)$$

Here c_s and y_s denote the nonstochastic steady-state values of private consumption and output.⁹

To assess the magnitude of the multiplier and the elasticities, we used measures of n_t , c_t , g_t , and y_t described in Christiano and Eichenbaum (1992). We then estimated n_s , c_s/y_s , and $\bar{g}/y_s$ by the sample averages of our measures of n_t , c_t/y_t , and g_t/y_t over the period from 1955:4 to 1983:4. These equal 320.2, 0.55, and 0.177, respectively. In addition, we set $N = 2,190$, the number of hours in a quarter. Substituting these parameter values into (14) and (15), we obtain

$$\frac{dy_s}{d\bar{g}} = 1.22, \quad \eta_{y,g} = \eta_{n,g} = 0.22. \quad (16)$$

Since $dy_s/d\bar{g}$ exceeds unity, it follows that there can be a nonstochastic steady-state analog to the Keynesian multiplier in the neoclassical growth model.

Aiyagari et al. (1990) also provide a counterexample to the claim of Barro (1981, 1987) that the contemporaneous output-government multiplier must be less than one. Specifically, using log-linear approximation to the decision rules, they show that when $\rho = 0.9999$, this multiplier equals 1.15.¹⁰ However, when

⁹For a derivation of (15) see Aiyagari et al. (1990, n. 14).

¹⁰The values for the structural parameters underlying these calculations as well as those pertaining to the interest rate response of changes in government consumption are $\beta = 1.03^{-0.25}$, $\theta = 0.347$, $\delta = 0.021$, $\gamma = 7.00$, $\lambda = 0.0047$, $\bar{g} = 199.0$. See Christiano and Eichenbaum (1992).

ρ equals 0.97, the point estimate obtained by Christiano and Eichenbaum (1992), the multiplier equals 0.78.

3.3 *The effects of shocks to government consumption on the risk-free interest rate*

In equilibrium, the time t risk-free rate of return on consumption loans, r_t , is equal to the expected value of the planner's intertemporal marginal rate of substitution on consumption. To calculate r_t , we use decision rules for k_t and n_t obtained using the value function iteration procedure described in Christiano (1990). This procedure involves discretizing the support of the exogenous processes of the model and of k_t . We assume that g_t^P is a realization from a three-state Markov chain with

$$g_t^P = \{g_1^P, g_2^P, g_3^P\} = \{-0.1444, 0, 0.1444\}, \quad (17)$$

$$\text{Prob}[g_{t+1}^P = g_j^P | g_t^P = g_i^P] = \pi_{ij}, \quad i, j, = 1, 2, 3,$$

$$\pi = \begin{bmatrix} 0.97310 & 0.0250 & 0.00190 \\ 0.00625 & 0.9875 & 0.00625 \\ 0.00190 & 0.0250 & 0.97310 \end{bmatrix}.$$

With this specification, g_t^P has the AR(1) specification given by (13) with $\rho = 0.97$. While μ_t is conditionally heteroscedastic, it has an unconditional standard deviation of 0.020, which corresponds to the value estimated in Christiano and Eichenbaum (1992).

We assume that g_t^T is iid and can take on the values -0.1444 , 0 , and 0.1444 with probabilities 0.00311, 0.99938, and 0.000311, respectively. With this specification, g_t^T has an unconditional mean of 0 and a standard deviation of 0.0036.¹¹ We assume that λ_t is also iid and takes on the values -0.02 , 0.0047 , and 0.0295 with probabilities 0.264, 0.471, and 0.264, respectively. It follows that λ_t has an unconditional mean equal to 0.0047 and a standard deviation of 0.081, which correspond to the values estimated in Christiano and Eichenbaum (1992). For the capital stock, we used a grid with 15,000 points evenly spaced between $k_t = 7,000$ and 23,300.

The response of r_t to a shock in either g_t^T or g_t^P depends on the values of the date t state variables, k_t , λ_t , g_t^P , g_t^T . We report the expected value of the

¹¹Under our assumptions, the univariate Wold representation of $\log g_t$ is given by $\log g_t = 0.97 \log g_{t-1} + c + \varepsilon_t - 0.03\varepsilon_{t-1}$, where c is a constant. This is very close to the specification in Christiano and Eichenbaum (1992) in which the moving-average coefficient in the representation for $\log g_t$ equals zero.

derivative of r_t with respect to g_t^P and g_t^T , which we denote by Δ_r^P and Δ_r^T , respectively. The expectation is taken with respect to the steady-state distribution of the capital stock and is evaluated at the mean values of the exogenous shocks. Performing this calculation, we find that $\Delta_r^P = 0.0048$ and $\Delta_r^T = 0.0005$. Thus, a persistent change of 1% in government consumption induces roughly a 0.5 basis point increase in the interest rate. In contrast, a transient change of 1% in government consumption induces a barely perceptible rise of 0.05 basis point.

The fact that $\Delta_r^P > 0$ shows that a persistent increase in government consumption does change the interest rate. The fact that $\Delta_r^P > \Delta_r^T$ provides a counterexample to claims in Hall (1980) and Barro (1981, 1987) that the interest rate necessarily responds more to a temporary change in government consumption than to a persistent change. Finally, we note that according to our model, the interest rate effects of movements in government consumption – temporary or persistent – are very small. This is the case even though the output and employment effects of changes in government consumption are substantial. From this perspective, it is not surprising that Barro (1981, 1987) fails to find markedly higher real interest rates during periods of temporarily high government consumption, such as wartime.

4. Concluding remarks

Our analysis of the effects of government consumption was conducted under several simplifying assumptions. First, we assumed that government consumption is financed by lump-sum taxes. This seems like a natural benchmark for isolating the theoretical effects of changes in government consumption on aggregate economic activity. Nevertheless, this is clearly not a natural assumption to make from the perspective of empirical work. Recent papers by Zhu (1990), Aiyagari (1991), and Chari et al. (1991) analyze the effects of changes in government consumption financed by distortionary taxes. We also assumed that agents' preferences are additively separable across public and private consumption. Several authors, including Barro (1981, 1987), Kormendi (1983), Aschauer (1985), Aiyagari (1988), and Christiano and Eichenbaum (1992) have considered models in which this assumption is relaxed. Numerical results in Christiano and Eichenbaum (1992) and theoretical results in Aiyagari (1988) indicate that the results of this paper can be extended to allow for certain forms of nonseparabilities between public and private consumption.

References

- Aiyagari, S. Rao, 1988, What are the output effects of government purchases?, Research Department working paper no. 411 (Federal Reserve Bank of Minneapolis, Minneapolis, MN).

- Aiyagari, S. Rao, 1991, Expansionary effects of fiscal policy in a simple real business cycle model of Markov consistent labor taxation, Research Department working paper no. 455 (Federal Reserve Bank of Minneapolis, Minneapolis, MN).
- Aiyagari, S. Rao, Lawrence J. Christiano, and Martin S. Eichenbaum, 1990, The output, employment, and interest rate effects of government consumption, NBER working paper no. 3330 (National Bureau of Economic Research, Cambridge, MA), also Research Department working paper no. 456 (Federal Reserve Bank of Minneapolis, Minneapolis, MN) and Institute for Empirical Macroeconomics discussion paper no. 25 (Federal Reserve Bank of Minneapolis and University of Minnesota, Minneapolis, MN).
- Aschauer, David A., 1985, Fiscal policy and aggregate demand, *American Economic Review* 75, 117–127.
- Barro, Robert J., 1981, Output effects of government purchases, *Journal of Political Economy* 89, 1086–1121.
- Barro, Robert J., 1987, *Macroeconomics*, 2nd ed. (Wiley, New York, NY).
- Barro, Robert J. and Robert G. King, 1984, Time-separable preferences and intertemporal-substitution models of business cycles, *Quarterly Journal of Economics* 99, 817–839.
- Baxter, Marianne and Robert G. King, 1988, Multipliers in equilibrium business cycles models, Manuscript (University of Rochester, Rochester, NY).
- Chari, V.V., Lawrence J. Christiano, and Patrick J. Kehoe, 1991, Optimal fiscal policy in a business cycle model, Research Department working paper no. 465 (Federal Reserve Bank of Minneapolis, Minneapolis, MN).
- Christiano, Lawrence J., 1990, Linear-quadratic approximation and value-function iteration: A comparison, *Journal of Business and Economic Statistics* 8, 99–113.
- Christiano, Lawrence J. and Martin S. Eichenbaum, 1992, Current real-business-cycle theories and aggregate labor market fluctuations, *American Economic Review* 82, 430–450.
- Hall, Robert E., 1980, Labor supply and aggregate fluctuations, in: *On the state of macroeconomics*, Carnegie-Rochester conference series on public policy, Vol. 12 (North-Holland, Amsterdam).
- Kormendi, Roger C., 1983, Government debt, government spending, and private sector behavior, *American Economic Review* 73, 994–1010.
- Zhu, Xiadong, 1990, Optimal fiscal policy in a stochastic growth model, Manuscript (Graduate School of Business, University of Chicago, Chicago, IL).